Chapter 2 : Specific Heat of Solids

The law of Dulong-Petit

As far back as 1819, however, it had also been know that for many solids the specific heat is given by

$$C = 3k_B \quad \text{per atom}$$

ou

$$C = 3R$$

Which is know as the *Law of Dulong-Petit*. While this law is not always corrects, if frequently is close to true. For example, at room temperature we have

Material	C/R
Aluminum	2.91
Antimony	3.03
Copper	2.94
Gold	3.05
Silver	2.99
Diamond	0.735

With the exception of diamond, the law seems to hold extremely well at room temperature, although at lower temperatures all materials start to deviate from this law, and typically C drops rapidly below some temperature.

Before Einstein, in 1896 Boltzmann constructed a model that accounted for the law of Dulong-Petit fairly well. In his model, each atom in the solids is bound to neighboring atoms. Focusing on a single particular atom, we imagine the atom as being in a harmonique well formed y the interaction its neighbours. In such a classical statistical mechanical model, specific heat of the vibration of the atom is $3k_B$ per atom, in agreement with Dulong-Petit.

1. Einstein's Calculation

Specific heat

$$C = 3k_B(\beta\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

For high temperature limit $k_B T \gg \hbar\omega \Rightarrow e^{\beta\hbar\omega} - 1 \sim \beta\hbar\omega$.

$$C \to 3k_B$$

This so that at the high temperature limit recover the law of Dulong-Petit.

For low temperature limit $k_B T \ll \hbar\omega$.

$$C \sim T$$

This so that at the low temperature limit the heat capacity per atom, the system gets stuck in only the ground state eigenstate, and the specific heat vanishes rapidly.

Related:

Einstein's Calculation

2. Debye's Calculation

Debye decided that the oscillation modes were waves with frequencies $\omega(\mathbf{k}) = v|\mathbf{k}|$ with v the sound velocity and for each $\mathbf{k}$ there should be three possible oscillation modes. Thus he wrote an expression entirely analogous to Einstein's expression

$$\langle E\rangle = 3\sum_k \hbar\omega(k)\left[n_B(\beta\hbar\omega(k)) + \frac{1}{2}\right]$$

Specific heat

$$C = Nk_B\left(\frac{k_B T}{\hbar\omega_d}\right)^3 \frac{12\pi^4}{5} \sim T^3$$

We know that according to law of Dulong-Petit, specific heat C not depend on T^3, this so that Debye's model has a problem. Debye understood that the problem with his approximation is that it allow an infinite number of sound wave modes- up to arbitrarily large $\mathbf{k}$. This would imply more sound wave modes that there are atoms in the entire system (why the number of sound wave can't bigger that the number of atom in system?). To fix this problem, Debye decided to nor consider sound wave above some maximum frequency ω_{cutoff} , with this frequency chosen such that there are exactly $3N$ sound wave modes in the system. We thus define ω_{cutoff} via

$$N = \int_0^{\omega_{\text{cutoff}}} d\omega g(\omega)$$

summary

- Specific heat of materials is due to atomic vibrations
- Boltzmann and Einstein models consider these vibrations as N simple harmonique oscillators.
- Boltzmann classical analysis obtains law of Dulong-Petit $C = 3Nk_B = 3R$
- Einstein quantum analysis shows that at temperature below the oscillator frequency, degrees of freedom freeze out, and heat capacity drops exponentially
- Debye Model treats oscillations as sound waves
    - $\omega = v|k|$, similar to light (but three polarizations not two)
    - quantization similar to Planck quantization of light

    - Maximum frequency cutoff ($\hbar\omega_{\text{Debye}} = k_B T_{\text{Debye}}$) necessary to obtain a total of only $3N$ degree of freedom
    - obtain Dulong-Petit at high T and $C \sim T^3$ at low T
- Metals have an additional (albeit small) linear T term in the specific heat

Related:

Professor Steven H. Simon.pdf
Note_2
Debye's Calculation
About Periodic
Method of reading scientific books.